

1a) fixed charge n/w flow prob

Parameters

- * T = number of months
- * d_t = demand in month t
- * S = setup cost if production happens in month t
- * C = unit production cost (same each month)
- * h = holding cost per unit per month
- * M = upper bound on production in any period

Variables

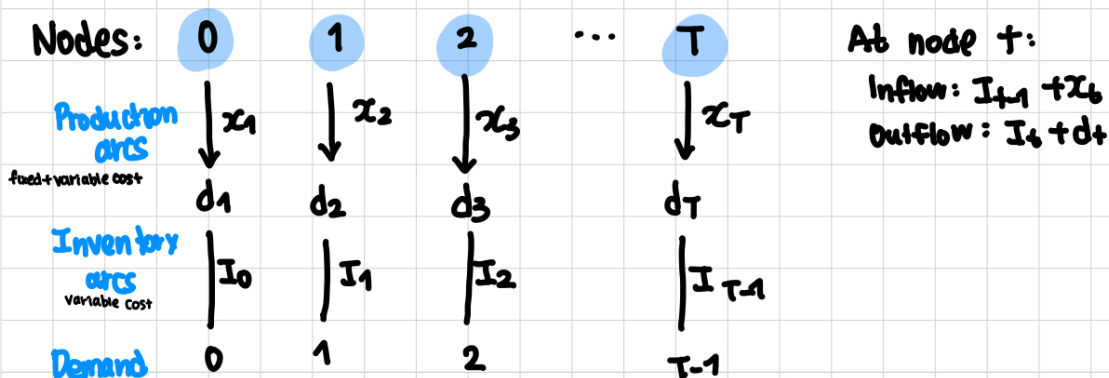
- * x_t = flow on production arc $(t-1, t)$
- * I_t = flow on inventory arc $(t, t+1)$
- * y_t = binary variable indicating production in period t $\begin{cases} 1 & \text{production} \\ 0 & \text{else} \end{cases}$

Mathematical Model

$$\min \sum_{t=1}^T (Cx_t + Sy_t) + \sum_{t=1}^{T-1} hI_t$$

- $I_{t+1} + x_t = d_t + I_t$ for all $t=1 \dots T$
- $x_t \leq My_t$ for all $t=1 \dots T$
- $I_0 = I_T = 0$
- $x_t \geq 0, I_t \geq 0$ for all t
- $y \in \{0,1\}$ for all t

Graph



1 b) Shortest Path

Key property

There exists an optimal solution where $I_{t-1} \cdot x_t = 0$ for all t

Implications:

- Production only occurs when inventory is zero
- Each production run covers demand for a consecutive block of periods until next prod.
- Solution decomposes into regeneration intervals

Graph Representation

Nodes: $0, 1, 2, \dots, T$

- Node i : start of period $i+1$
- Node 0: beginning
- Node T : end

Arcs: (i, j) for $i < j$

- Produce in period $i+1$ to cover all demand from period $i+1$ to j
- next production will occur in period $j+1$

Cost of arc (i, j)

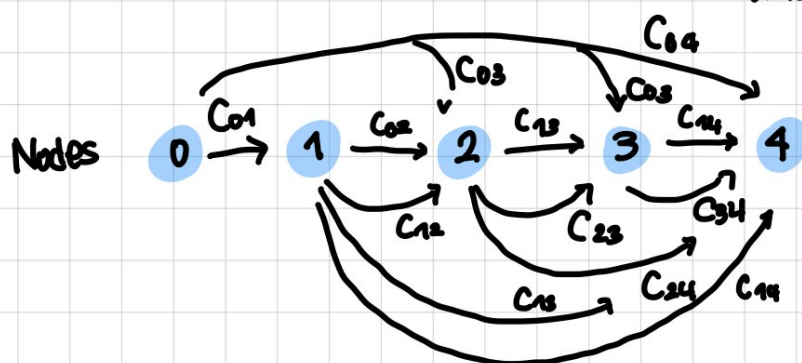
- S : setup cost
- Production cost: $c \cdot x (d_{i+1} + d_{i+2} + \dots + d_j)$
- Holding cost

$$\therefore \text{Cost}(i, j) = S + c \sum_{k=i+1}^j d_k + h \sum_{k=i+2}^j (k - (i+1)) d_k$$

units for period $i+2$: held 1 mo. $\rightarrow h \cdot d_{i+2} \cdot 1$

units for period $i+3$: held 2 mo. $\rightarrow h \cdot d_{i+3} \cdot 2$

units for period j : held $j - (i+1)$ mo $\rightarrow h \cdot d_j \cdot (j - (i+1))$



where C_{ij} = cost of arc (i, j)

Mathematical Formulation

Let w_{ij} = cost of arc (i, j)

Decision Variable $z_{ij} \begin{cases} 1 & \text{arc } (i, j) \text{ used in path} \\ 0 & \text{else} \end{cases}$

Shortest Path

$$\min \sum_{i=0}^{T-1} \sum_{j=i+1}^T w_{ij} z_{ij}$$

subject to

$$\sum_{j=1}^T z_{0j} = 1 \quad (\text{start at node 0})$$

$$\sum_{j=i+1}^T z_{ij} - \sum_{k=0}^i z_{ik} = 0 \quad \forall i=1, \dots, T-1$$

$$\sum_{i=0}^{T-1} z_{iT} = 1 \quad (\text{end at node } T) \quad z_{ij} \in \{0, 1\} \quad \forall i, j$$

2a) ILP

$i=1, \dots, m$ (team members)
 $j=1, \dots, n$ (responsibility)

Parameters

C_{ij} : cost of assigning team member i to responsibility j

Decision Var.

$x_{ij} \begin{cases} 1 & \text{team member } i \text{ is assigned to resp. } j \\ 0 & \text{else} \end{cases}$

$$\rightarrow \min \sum_{i=1}^m \sum_{j=1}^n C_{ij} x_{ij}$$

subject to

$$\sum_{i=1}^m x_{ij} = 1 \quad \text{for all } j=1, \dots, n \quad \dots \textcircled{1}$$

$$\sum_{j=1}^n x_{ij} \leq 1 \quad \text{for all } i=1, \dots, m \quad \dots \textcircled{2}$$

$$x_{ij} \geq 0, \quad x_{ij} \in \mathbb{Z}$$

Special cases

1. if $m=n$

2. if $m > n$

3. if $m < n$

Change $\textcircled{2} \rightarrow \sum_{j=1}^n x_{ij} = 1$ Some team members unassigned Problem infeasible

2 b) Proof

Write in matrix form $Ax=b$

- A has $m+n$ rows
- Row i (for $i=1, \dots, m$) $\rightarrow \sum_j x_{ij} \leq 1$
- Row $m+j$ (for $j=1, \dots, n$) $\rightarrow \sum_i x_{ij} = 1$
- each variable x_{ij} appears in exactly 2 constraints: row i , row $m+j$

so each column of A has exactly two 1s, and those 1s are in different blocks

$\rightarrow$ Totally Unimodular

Replace $x_{ij} \in \{0,1\}$ with $x_{ij} \geq 0$

The LP is $\min \{C^T x : Ax=b, x \geq 0\}$

if A is TU and b integer, then every basic feasible solution is integer.

The optimal LP solution occurs at a basic feasible solution
With above, it is integer $\therefore$ satisfies the original ILP's binary constraints
 $\therefore$ LP relaxation solves the ILP exactly.

LP Relaxation (Root Node)

Constraints

$$1. -8x_1 + 16x_2 \leq 20 \rightarrow x_2 \leq 0.5x_1 + 1.25$$

$$2. 6x_1 + 10x_2 \leq 52 \rightarrow x_2 \leq 5.2 - 0.6x_1$$

$$3. x_1 \leq 5$$

$$\Rightarrow 0.5x_1 + 1.25 = 5.2 - 0.6x_1$$

$$1.1x_1 = 3.95$$

$$\therefore x_1 \approx 3.59$$

$$x_2 = 0.5(3.59) + 1.25 \approx 3.045$$

$$Z_{LP} = -3.59 + 3 \cdot 3.045 \approx 5.45$$

Node 0
 $x = (3.59, 3.045)$
 $Z = 5.45$

Non integer $\rightarrow$ need to branch.

Branching on x_2

Node 1: $x_1 \leq 3$

Node 2: $x_1 \geq 4$

Node 1: insert $x_1 = 3$

$$-24 + 16x_2 \leq 20 \Rightarrow 16x_2 \leq 44 \therefore x_2 \leq 2.75$$

$$18 + 10x_2 \leq 52 \Rightarrow 10x_2 \leq 34 \therefore x_2 \leq 3.4$$

$$Z = -3 + 3x_2$$

Choose $x_2 = 2.75$

$$-24 + 44 = 20 \checkmark$$

$$18 + 27.5 = 45.5 \leq 52 \checkmark$$

$$x_1 \leq 5 \checkmark$$

$$x_1 \leq 3 \checkmark$$

$$Z = -3 + 9.25 = 5.25$$

Non-integer

Node 2: $x_1 \geq 4 \rightarrow$ insert $x_1 = 4$

$$-32 + 16x_2 \leq 20 \Rightarrow x_2 \leq 3.25$$

$$24 + 10x_2 \leq 52 \Rightarrow x_2 \leq 2.8$$

$$Z = -4 + 3x_2 \therefore x_2 \leq 2.8$$

$$Z = -4 + 8.4 = 4.4$$

$$-32 + 44.8 = 12.8 \leq 20 \checkmark$$

Check if increasing x_1 helps:

$$x_1 = 5, x_2 = 2.2, Z = -5 + 6.6 = 1.6 \nrightarrow$$

$$x_1 = 4.5, x_2 = 2.7 + 10x_2 \leq 52 \Rightarrow 10x_2 \leq 25 \therefore x_2 \leq 2.5$$

$$Z = -4.5 + 7.5 = 3 \therefore \nrightarrow$$

So for node 2, $x = (4, 2.8)$ $Z = 4.4$ is best

$\rightarrow$ Both non-integer

$\rightarrow$ repeat this until int.

